

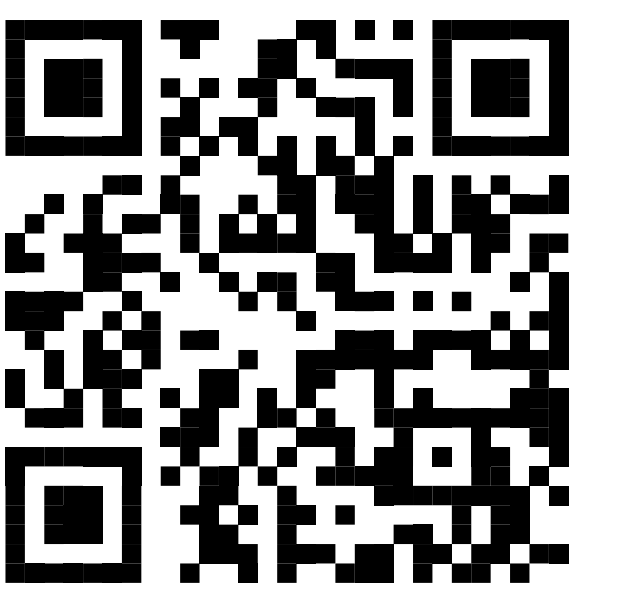


DexGraspNet: A Large-Scale Robotic Dexterous Grasp Dataset for General Objects Based on Simulation

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Project Page & Code

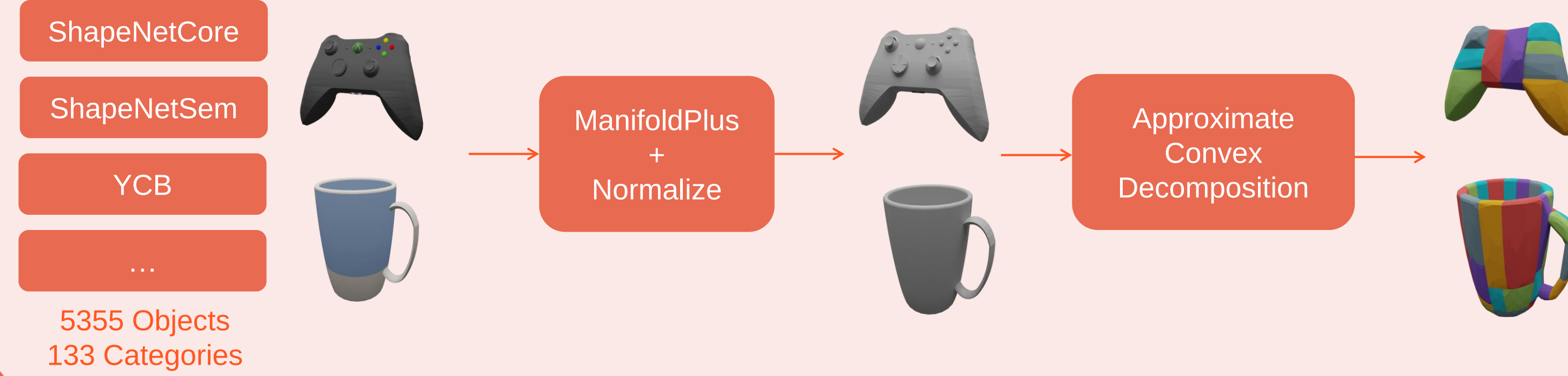
Outstanding Manipulation Paper Finalist

Introduction

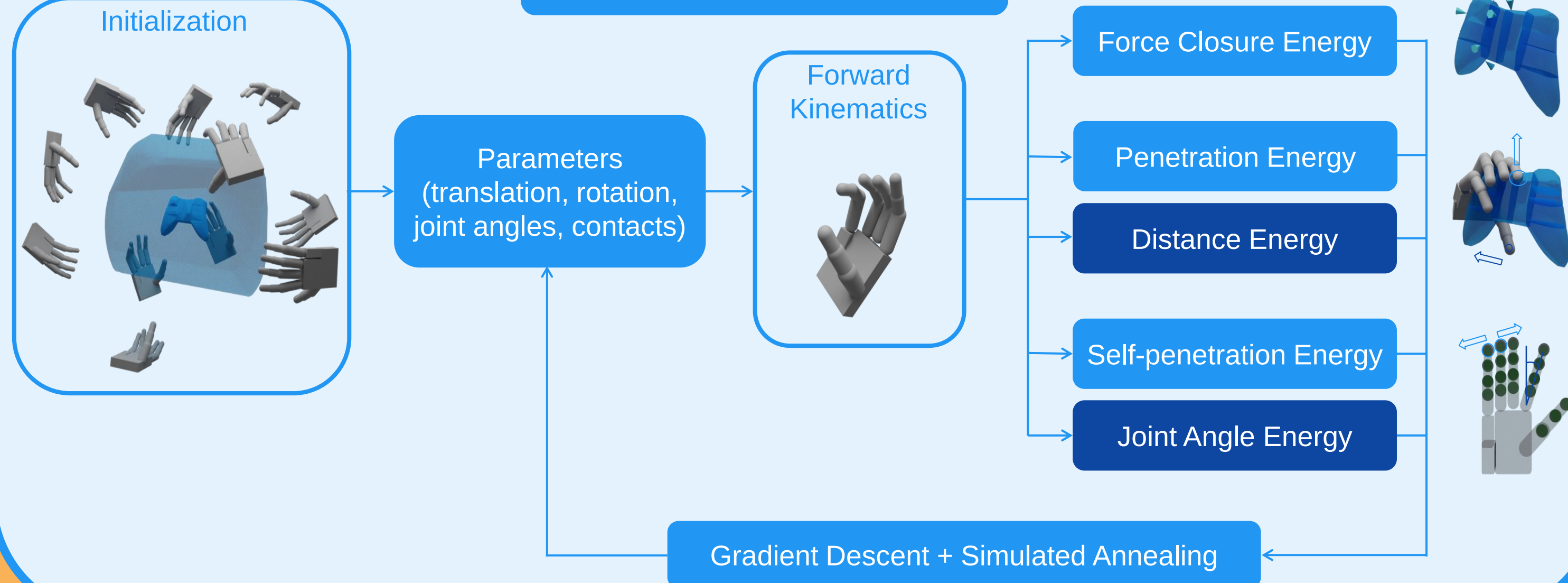
Robotic dexterous grasping is crucial for human-like manipulation. But it is highly under-explored due to the lack of a large-scale dataset. We present **DexGraspNet**, a large-scale robotic dexterous grasping dataset consisting of 1.32M grasps, 5355 objects, and 133 object categories, generated by our **efficient synthesis pipeline**. Although the dataset is generated for ShadowHand, our pipeline is customizable to other hands and settings. We provide the **implementations for ShadowHand, Allegro, MANO, and the table-top scenario**.

Method

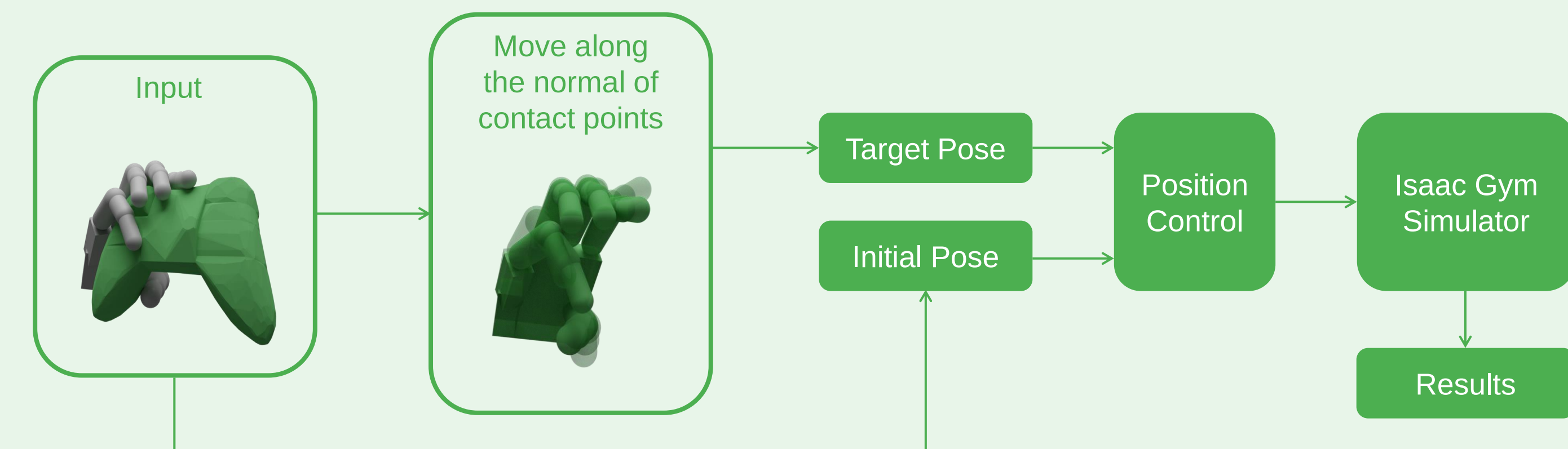
Asset Processing



Grasp Synthesis



Grasp Validation



Visualization of DexGraspNet



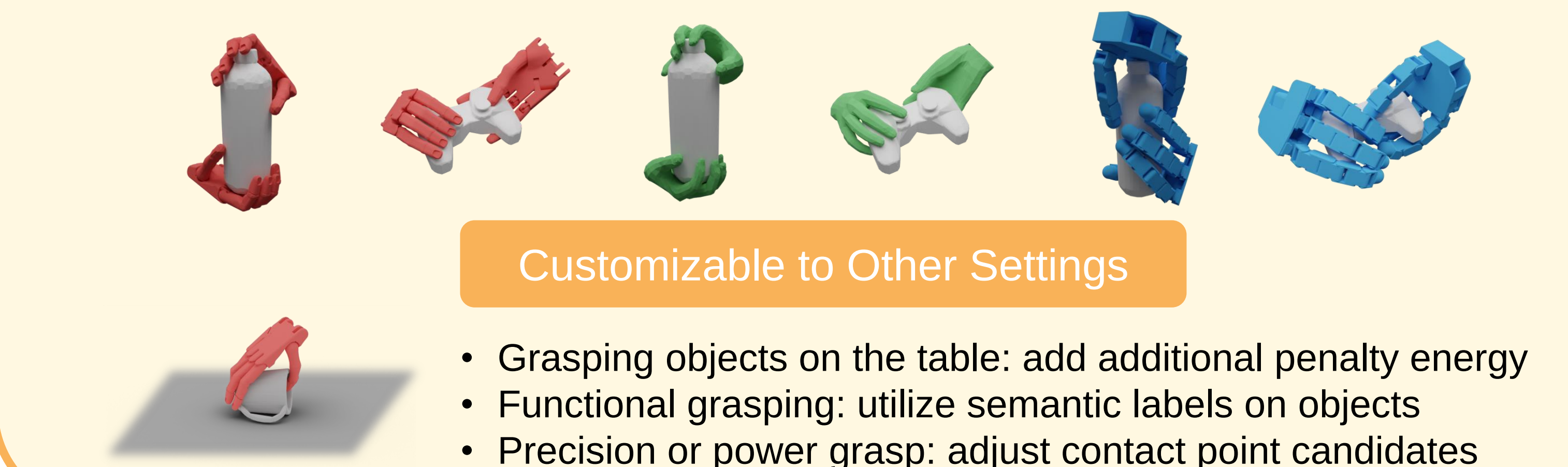
Qualitative Results

Diverse Grasp Synthesis



Comparison with Graspl!

Customizable to Other Hands



Customizable to Other Settings

- Grasping objects on the table: add additional penalty energy
- Functional grasping: utilize semantic labels on objects
- Precision or power grasp: adjust contact point candidates

Quantitative Comparison and Benchmark

Dexterous Grasp Dataset

Dataset	Hand	Observations	Sim./Real	Grasps	Obj.(Cat.)	Grasps per Obj.	Method
ObMan [14]	MANO	-	Sim.	27k	2772(8)	10	Graspl! Estimation
HO3D [15]	MANO	RGBD	Real	77k	10	>7k	Human annotation
DexYCB [16]	MANO	RGBD	Real	582K	20	>29k	Capture
ContactDB [17]	MANO	RGBD+thermal	Real	3750	50	75	Capture
ContactPose [18]	MANO	RGBD	Real	2306	25	92	Graspl! Optimization
DDGdata [9]	ShadowHand	-	Sim.	6.9k	565	>100	
DexGraspNet (Ours)	ShadowHand	-	Sim.	1.32M	5355(133)	>200	

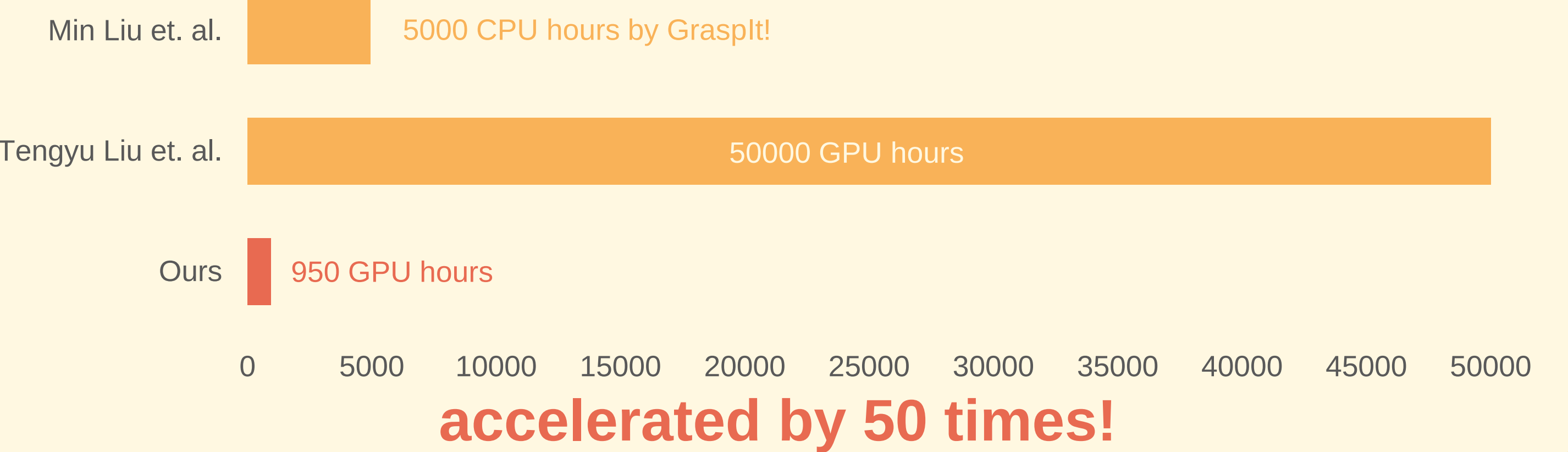
Benchmark

Method (Training Dataset)	Tested on DexGraspNet			Tested on DDGdata		
	success ↑	Q_1 ↑	pen ↓	success ↑	Q_1 ↑	pen ↓
DDG (DDGdata)	57.4	0.0493	0.353	56.4	0.0461	0.333
DDG (DexGraspNet)	67.5	0.0582	0.173	75.9	0.0524	0.134
GraspTTA (DDGdata)	17.1	0.0126	0.720	23.7	0.0265	0.666
GraspTTA (DexGraspNet)	24.5	0.0271	0.678	39.3	0.0790	0.547

Method (Training Dataset)	DexGraspNet		DDGdata	
	H mean	H std	H mean	H std
DDG (DDGdata)	4.958	2.653	3.709	1.942
DDG (DexGraspNet)	5.683	1.993	4.272	1.287
GraspTTA (DDGdata)	5.952	0.934	5.837	1.047
GraspTTA (DexGraspNet)	6.111	0.569	5.947	0.528

Synthesis Acceleration

Time to Generate 1.32M Grasps (Hours)



Conclusions

Contribution I: DexGraspNet

- **Largest** dexterous grasping dataset
 - 1.32M grasps, 5355 objects, 133 categories
- **Diverse** objects and grasps

Contribution II: A complete pipeline

- **Robust** to low quality object meshes
- **Efficient** optimization method
- **Customizable** to other hands/settings